People's Democratic Republic of Algeria

Ministry of Higher Education and Scientific Research

Higher School of Computer Science
8 May 1945 - Sidi Bel Abbes

MASTER'S THESIS

In partial fulfillment of the requirements for the Master's degree

Field: **Computer Science**

Specialty: **Ingénierie des Systèmes Informatiques & Intelligence Artificielle et Science de Données**

Theme

Design and Deployment of a Digital Twin for the Optimization of an Intelligent Irrigation System in Precision Agriculture

---

Presented by:

- Mr. Lebaili Mohamed Ouail
- Mr. Senhadji Mohamed Said

Defended on: 07/07/2026	**Before the jury composed of:**
- Dr. BENCHERIF Khayra	Présidente
- Dr. BEZZAOUCHA Fatima Souad	Examinatrice
- Pr. MALKI Mimoun	Encadreur
- Dr. MALKI Abdelhamid	Co-Encadreur
- Dr. MALKI Abdelhamid	Représentant de l'incubateur

Academic Year: 2025 / 2026

Dedication

To our families, teachers, and everyone who supported us throughout this final-year project. This work is dedicated to those who believe that engineering can serve sustainability, agriculture, and society.

Acknowledgements

We would like to express our sincere gratitude to our supervisors, teachers, and the staff of École Supérieure en Informatique -08 Mai 1945- Sidi Bel Abbès for their guidance, scientific advice, and support during the realization of this project. We also thank the laboratory team who made the smart farming prototype, Libelium Smart Agriculture kit, XBee communication infrastructure, Raspberry Pi gateway, and experimental environment available for the implementation and validation of this work.

This thesis also benefited from recent research on Digital Twins, smart irrigation, precision agriculture, Internet of Things, machine learning, and sustainable agricultural water management. These works allowed us to position the proposed platform not only as a software prototype, but also as a research-driven engineering contribution.

Abstract

Agriculture is increasingly affected by water scarcity, climate variability, soil degradation, and the growing demand for food production. Irrigation management is one of the most critical factors in this context, because traditional irrigation practices are often based on fixed schedules, manual observation, or empirical experience. Such approaches can lead to water waste, plant stress, and reduced productivity. In response, this thesis presents the design and deployment of Agrio, a Digital Twin-based intelligent irrigation platform for precision agriculture.

The proposed system integrates a physical smart farming prototype, Libelium Smart Agriculture sensing equipment, XBee radio-frequency communication, a Raspberry Pi gateway, HiveMQ/MQTT messaging, a FastAPI backend, a PostgreSQL database, a Random Forest-based machine learning service, a Digital Twin state and simulation layer, a Next.js web dashboard, a Flutter mobile application, a WeMos actuator, and a water pump. The platform receives real sensor data from a controlled prototype environment, normalizes and stores the measurements, updates the virtual state of the irrigation zone, predicts irrigation need, simulates future moisture evolution, generates recommendations, supports human validation, schedules irrigation actions, and controls the physical pump through MQTT commands.

The Master 2 contribution of this work is the scientific study of Digital Twins in agriculture, with a focus on their maturity levels, enabling technologies, irrigation applications, and limitations. The engineering contribution is the implementation of an end-to-end cyber-physical prototype that connects sensing, communication, backend processing, artificial intelligence, simulation, user interfaces, deployment, and actuation. The results show that the system goes beyond a classical IoT monitoring dashboard by providing synchronization, prediction, what-if simulation, recommendation, scheduling, and physical control. The current implementation remains a prototype tested in a controlled environment, and future work should extend it to long-term field validation, stronger agronomic calibration, cybersecurity, and advanced optimization techniques.

Keywords: Digital Twin, precision agriculture, smart irrigation, IoT, MQTT, XBee, Raspberry Pi, FastAPI, PostgreSQL, machine learning, Random Forest, water management, WeMos, smart farming.

Résumé

L'agriculture moderne fait face à des défis importants liés à la rareté de l'eau, aux variations climatiques et à la nécessité d'optimiser la production agricole. Dans ce contexte, l'irrigation intelligente constitue une application centrale de l'agriculture de précision. Ce mémoire présente la conception et le déploiement d'Agrio, une plateforme d'irrigation intelligente basée sur le concept de jumeau numérique. Le système proposé combine un prototype physique, un kit Libelium Smart Agriculture, une communication radiofréquence XBee, une passerelle Raspberry Pi, la messagerie MQTT/HiveMQ, un backend FastAPI, une base de données PostgreSQL, un service d'apprentissage automatique basé sur Random Forest, une couche de simulation du jumeau numérique, un tableau de bord web, une application mobile Flutter, un actionneur WeMos et une pompe d'eau.

La contribution scientifique porte sur l'étude de l'état de l'art des jumeaux numériques en agriculture et leur positionnement par rapport aux systèmes d'irrigation intelligents existants. La contribution d'ingénierie réside dans la mise en œuvre d'une chaîne complète reliant les capteurs, la communication IoT, le traitement backend, la prédiction, la simulation, la recommandation, la validation humaine et le contrôle physique de l'irrigation. Les résultats montrent que le système dépasse la simple visualisation IoT en intégrant la synchronisation, la prédiction, la simulation et l'action.

Mots-clés : jumeau numérique, agriculture de précision, irrigation intelligente, IoT, MQTT, apprentissage automatique, FastAPI, PostgreSQL.

ملخص

تواجه الزراعة الحديثة تحديات كبيرة مرتبطة بندرة المياه والتغيرات المناخية والحاجة إلى تحسين الإنتاجية. في هذا السياق، تمثل أنظمة الري الذكي أحد أهم تطبيقات الزراعة الدقيقة. يقدم هذا العمل منصة Agrio، وهي منصة ري ذكية تعتمد على مفهوم التوأم الرقمي، حيث تجمع بين نموذج فيزيائي حقيقي، حساسات زراعية، اتصال لاسلكي، بوابة Raspberry Pi، بروتوكول MQTT، خادم خلفي FastAPI، قاعدة بيانات PostgreSQL، خدمة تعلم آلي، محاكاة رقمية، واجهة ويب، تطبيق هاتف، ومتحكم WeMos للتحكم في مضخة الماء.

تتمثل المساهمة العلمية في دراسة حالة الفن للتوائم الرقمية في الزراعة ومقارنة الأعمال الحالية، بينما تتمثل المساهمة الهندسية في بناء سلسلة كاملة من جمع البيانات إلى التنبؤ والمحاكاة والتوصية والتحكم الفعلي في المضخة. يبين العمل أن النظام لا يقتصر على مراقبة البيانات، بل يوفر تمثيلاً رقمياً متزامناً ودعماً ذكياً لاتخاذ القرار.

List of Abbreviations

Table 1: Abbreviations used in the thesis

Abbreviation	Meaning
AI	Artificial Intelligence
ADT	Agricultural Digital Twin
API	Application Programming Interface
CIS	Computer Systems Engineering
AIDS	Artificial Intelligence and Data Science
CPS	Cyber-Physical System
DB	Database
DSS	Decision Support System
DT	Digital Twin
ET	Evapotranspiration
ETo	Reference Evapotranspiration
FAO	Food and Agriculture Organization
FMIS	Farm Management Information System
IoT	Internet of Things
JSON	JavaScript Object Notation
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
REST	Representational State Transfer
RF	Radio Frequency
RDI	Regulated Deficit Irrigation
RFID	Radio-Frequency Identification
SBA	Sidi Bel Abbès
SQL	Structured Query Language
SWB	Soil Water Balance
UAV	Unmanned Aerial Vehicle
UI	User Interface
WSN	Wireless Sensor Network

Contents

List of Figures

List of Tables

Part I

Introduction

1 Context and Problem Statement

Agriculture is one of the foundations of food security, economic development, and social stability. However, it is also one of the sectors most exposed to climate variability, water scarcity, soil degradation, and increasing production pressure. In many regions, irrigation remains a central factor determining crop quality and yield. When irrigation is applied too late, plants may suffer water stress and productivity may decrease. When irrigation is applied too early or in excessive quantities, water is wasted, soil nutrients may leach, and the cost of operation increases. The challenge is therefore not simply to irrigate, but to irrigate at the right time, with the right quantity, and according to the actual state of the soil, crop, and environment.

Traditional irrigation practices are often based on fixed schedules, manual field observation, or the farmer's experience. These practices remain valuable because farmers possess contextual knowledge of their crops and land. Nevertheless, they can become insufficient when environmental conditions change quickly or when water availability becomes uncertain. Precision agriculture responds to this issue by relying on sensors, communication systems, data processing, and intelligent decision support. Smart irrigation is one of the most important applications of precision agriculture because it directly addresses water-use efficiency and sustainability.

Digital Twin technology offers a more integrated approach. A Digital Twin is not only a dashboard or a static simulation. It is a virtual representation of a physical system, continuously updated using real data, able to simulate future behavior, support decision-making, and potentially influence the physical system through feedback or actuation. In agriculture, Digital Twins can represent farms, plots, irrigation zones, greenhouses, plants, soil profiles, livestock, or agricultural machinery. Their value lies in connecting physical measurements, computational models, prediction algorithms, and operational decisions within a coherent cyber-physical loop.

This Master Diploma report focuses on the scientific background of precision agriculture, smart irrigation, Digital Twins, IoT, and artificial intelligence in agriculture. Its purpose is to study the foundations of these domains, define the concepts and maturity levels of Digital Twins, review the state of the art of agricultural Digital Twins and intelligent irrigation, and synthesize the main research gaps reported in the literature.

2 Objectives

The central research question of this Master report is: how can the literature on Digital Twins, IoT-based smart irrigation, and machine learning irrigation systems be analyzed in order to build a coherent scientific background for intelligent irrigation in agriculture? This question is divided into secondary questions: what are the main concepts and maturity levels of Digital Twins in agriculture? Which technologies are commonly used in IoT-based smart irrigation systems? How can machine learning support irrigation decision-making? What are the strengths and limitations of current research? Which criteria can be used to compare existing works in this domain?

The objectives of the Master Diploma report are: to study Digital Twins and smart irrigation systems in the agricultural domain; to present the fundamental concepts of precision agriculture, IoT, artificial intelligence, and simulation; to review existing research on agricultural Digital Twins and irrigation systems; to compare related works according to clear scientific criteria; and to identify recurring gaps, limitations, and research opportunities.

3 Thesis Outline

The report is organized according to the Master Diploma structure. The front matter contains the acknowledgements, abstracts, and lists. The introduction presents the context, problem statement, objectives, and outline. The background part introduces precision agriculture, smart irrigation, Digital Twins, IoT, and artificial intelligence. The state-of-the-art part reviews Digital Twins in agriculture and irrigation. The synthesis part compares existing works and extracts the main scientific lessons from the literature. The report ends with a general conclusion.

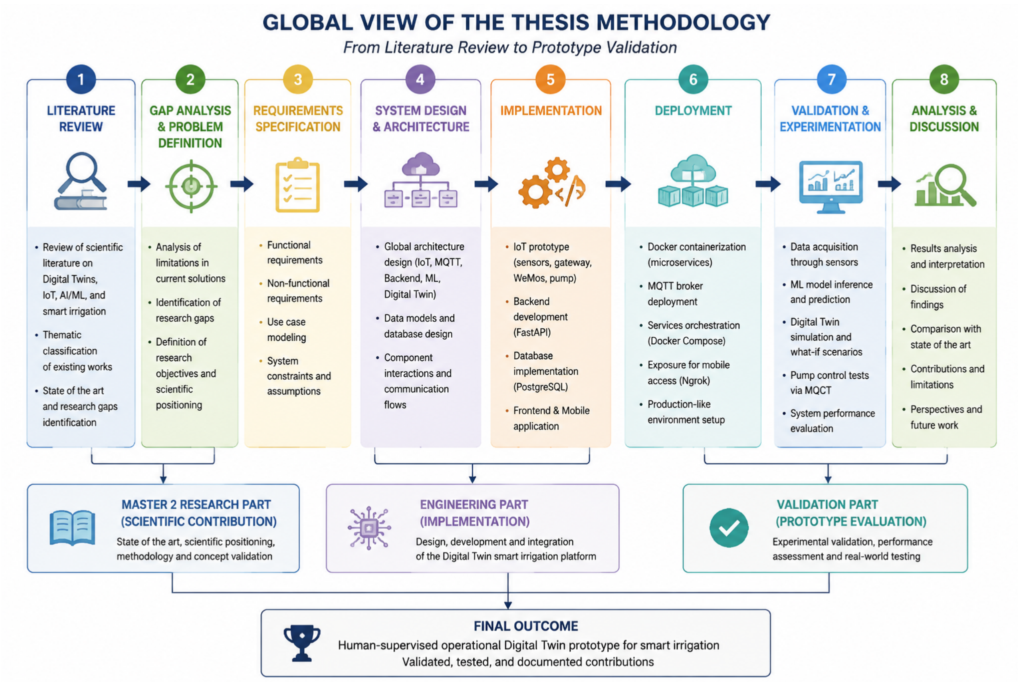

Figure 1: Global view of the thesis methodology from background study to literature synthesis. Source: Authors.

Table 2: Summary of Master Diploma objectives and contributions

Dimension	Objective	Contribution
Scientific background	Study Digital Twins, smart irrigation, IoT, and ML concepts	Structured theoretical foundation

Dimension	Objective	Contribution
State of the art	Analyze recent research on agricultural Digital Twins and smart irrigation	Literature review and research gap identification
Synthesis	Compare existing systems according to Digital Twin maturity and irrigation functionality	Research gap identification and literature synthesis

Part II

Background

Chapter 1

Precision Agriculture and Smart Irrigation

1.1 Introduction

The background of this thesis begins with precision agriculture and smart irrigation because they define the practical domain in which sensing, communication, data analysis, and control are applied. The purpose of this chapter is not to present a specific implementation, but to introduce the scientific concepts that explain why irrigation can be improved through digital technologies. The chapter therefore focuses on general definitions, common system components, measured variables, communication choices, and challenges reported in the literature.

Precision agriculture is commonly described as a data-driven management strategy. The International Society of Precision Agriculture defines it around the collection and analysis of temporal, spatial, and individual plant or animal data in order to support management decisions according to variability [1]. In irrigation, this idea means that water application should not be based only on fixed schedules. It should also consider soil moisture, crop water demand, weather conditions, irrigation method, field variability, and operational constraints.

Smart irrigation is one of the most important applications of precision agriculture. Garcia et al. [2] review IoT-based irrigation systems and show that most systems combine sensors, wireless communication, processing nodes, user interfaces, and irrigation actuators. In this literature, the main objective is to use measurements and decision rules, or predictive models, to apply water more efficiently while reducing waste and avoiding plant stress.

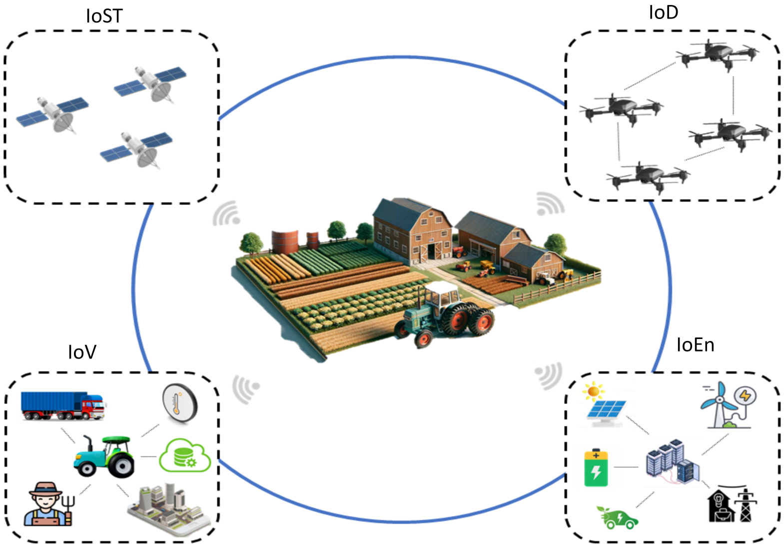

Figure 1.1: Integration of connected technologies in precision agriculture. Source: https://arxiv.org/abs/2404.06341.

1.2 Definition of precision agriculture

Precision agriculture is based on the observation that agricultural systems are not uniform. Soil texture, organic matter, salinity, water retention, crop stage, disease pressure, and microclimate may vary across a field and over time. A uniform management action can therefore be excessive in one zone and insufficient in another. Precision agriculture tries to reduce this mismatch by linking measurement, interpretation, and action.

From the definition used by ISPA [1], three ideas are central. First, precision agriculture is a management strategy, not only a collection of sensors. Second, it depends on data that can describe temporal and spatial variability. Third, its purpose is to support decisions that improve resource efficiency, productivity, quality, profitability, and sustainability. In irrigation studies, the same logic is applied to water: the system should identify when, where, and how much water is needed.

Table 1.1: Core concepts in precision agriculture

Concept	Meaning in the literature	Irrigation implication
Spatial variability	Differences between zones of the same field	Irrigation depth may differ by zone
Temporal variability	Changes over hours, days, seasons, and crop stages	Irrigation timing should adapt to current conditions
Decision support	Data are transformed into management recommendations	The system can recommend, schedule, or prioritize watering
Resource efficiency	Inputs are applied according to estimated need	Water waste and energy use can be reduced

Concept	Meaning in the literature	Irrigation implication
Sustainability	Production goals are balanced with environmental constraints	Over-irrigation, leaching, and salinity risks are considered

1.3 Irrigation management and water scarcity

Irrigation management is the process of deciding when to irrigate, how much water to apply, and how to deliver it. Classical irrigation scheduling may rely on farmer experience, fixed calendars, crop coefficients, evapotranspiration estimation, soil water balance, or manual observation. These methods remain useful, but they can be limited when weather conditions change rapidly or when field variability is high.

Water scarcity gives smart irrigation its practical importance. Garcia et al. [2] explain that agriculture is strongly affected by water availability and that research has increasingly focused on reducing water use in irrigation processes. Smart irrigation responds to this problem by combining measurements from the field with decision rules, models, or automation.

The main agronomic risk is not only water shortage. Under-irrigation can lead to plant water stress and yield reduction, while over-irrigation can waste water, increase pumping energy, move nutrients below the root zone, and contribute to salinity or drainage problems. A good irrigation decision must therefore balance crop demand, soil water storage, weather conditions, and system constraints.

1.4 Definition of smart irrigation

Smart irrigation can be defined as an irrigation approach that uses sensing, communication, data processing, and decision support to adjust irrigation according to measured or predicted needs. In IoT-based studies, the system usually contains field sensors, an embedded node or gateway, wireless communication, a database or cloud service, a decision layer, a user interface, and actuators such as pumps or valves [2, 3].

The distinction between conventional irrigation and smart irrigation is the role of data. In conventional irrigation, decisions are often periodic or manually triggered. In smart irrigation, decisions are linked to variables such as soil moisture, temperature, humidity, rainfall, crop stage, and evapotranspiration. Some systems remain advisory, while others close the loop by controlling actuators automatically.

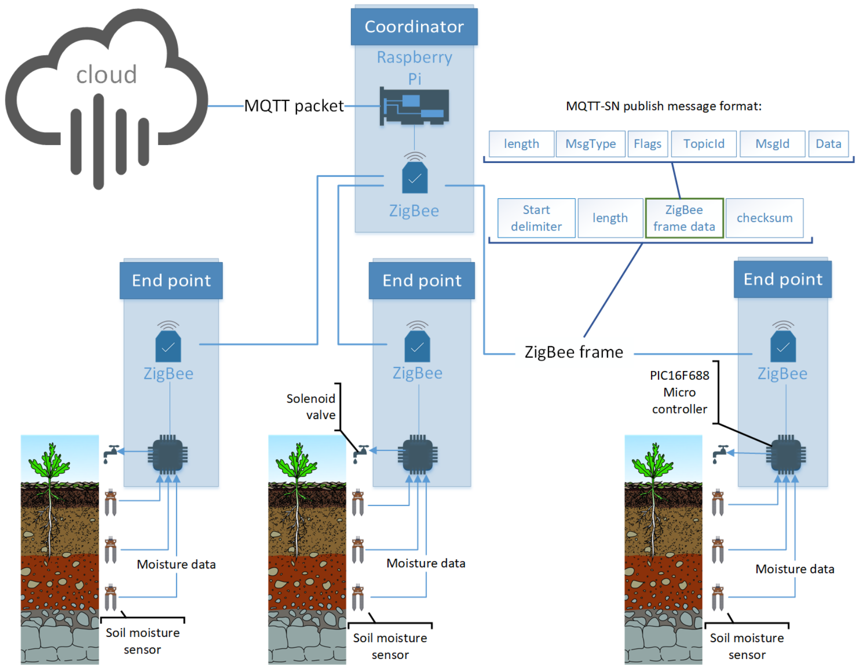

Figure 1.2: IoT-based smart irrigation architecture with endpoint, coordinator, and cloud server. Source: https://arxiv.org/abs/2009.12747.

Table 1.2: Levels of irrigation decision support

Level	Decision basis	Typical limitation
Manual irrigation	Farmer observation and experience	Depends on availability and subjective assessment
Scheduled irrigation	Calendar, fixed duration, or crop-stage rule	Does not react to short-term soil/weather changes
Threshold irrigation	Soil moisture or weather threshold	Simple and interpretable, but may ignore forecasts
Model-based irrigation	Soil water balance, evapotranspiration, or crop model	Requires calibration and reliable input data
AI-assisted irrigation	Machine learning prediction or optimization	Needs validated datasets and explainable outputs
Closed-loop control	Sensors and actuators connected through feedback	Requires safety, reliability, and fail-safe design

1.5 Measured variables in smart irrigation

The quality of a smart irrigation decision depends on the quality and relevance of the measured variables. Soil moisture is the most direct indicator because it describes water availability in the root zone. Garcia et al. [2] report that soil monitoring is one of the dominant categories in IoT irrigation systems, and soil moisture is the most common soil variable. Weather variables are also important because they affect evapotranspiration and future water demand.

Soil temperature, air temperature, relative humidity, rainfall, wind speed, solar radiation or light intensity, leaf wetness, pH, and electrical conductivity may also appear in the literature depending on the crop and objective. Not all systems need all variables. A low-cost system may use only soil moisture and temperature, while a research-grade system may include weather stations, remote sensing, and plant-based measurements.

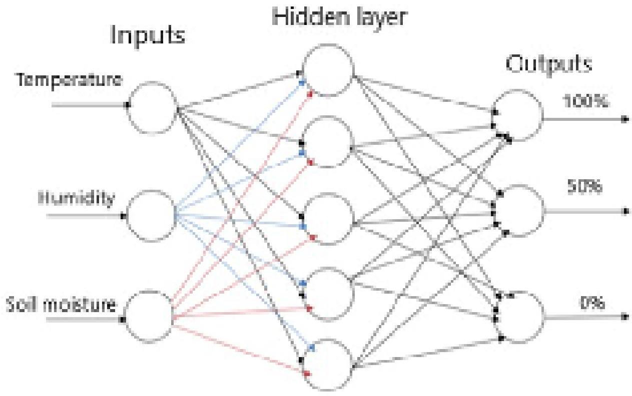

Figure 1.3: Measured variables and decision output in a smart irrigation controller. Source: https://arxiv.org/abs/2005.04158.

Table 1.3: Common variables used in smart irrigation literature

Variable group	Examples	Role in irrigation
Soil	Moisture, temperature, pH, electrical conductivity	Root-zone water status and soil condition
Weather	Air temperature, humidity, rainfall, wind, radiation	Evapotranspiration and short-term water demand
Water	Flow, level, salinity, turbidity	Availability and quality of irrigation water
Plant	Leaf wetness, canopy temperature, plant height	Plant response and stress indication
System state	Valve state, pump state, pressure, energy use	Execution monitoring and fault detection

1.6 Communication technologies in smart irrigation

Communication links connect the physical field to the digital decision layer. The literature reports Wi-Fi, Bluetooth, ZigBee, XBee, LoRa, LoRaWAN, GSM/GPRS, NB-IoT, 4G/5G, and wired links. The choice depends on range, energy consumption, data rate, cost, reliability, infrastructure availability, and the size of the agricultural area.

Garcia et al. [2] show that Wi-Fi, GSM/GPRS, ZigBee, Bluetooth, LoRa, and MQTT-based solutions appear in IoT irrigation studies. Long-range technologies such as LoRa are useful for remote fields with low data rates, while Wi-Fi is easier for small farms or greenhouse environments with existing network coverage. MQTT is often used as an application-layer protocol because its publish/subscribe structure decouples sensors, services, dashboards, and actuators.

Table 1.4: Communication technologies reported for IoT irrigation systems

Technology	Typical strength	Typical constraint
Wi-Fi	Low cost and easy integration	Shorter range and higher energy use
ZigBee/XBee	Low power mesh networking	Limited range per hop and lower data rate
LoRa/LoRaWAN	Long range and low energy	Low bandwidth and gateway planning
GSM/GPRS/4G/5G	Wide-area connectivity	Subscription cost and coverage dependency
Bluetooth	Simple short-range communication	Not suitable for large fields
MQTT	Lightweight publish/subscribe messaging	Requires broker availability and topic design

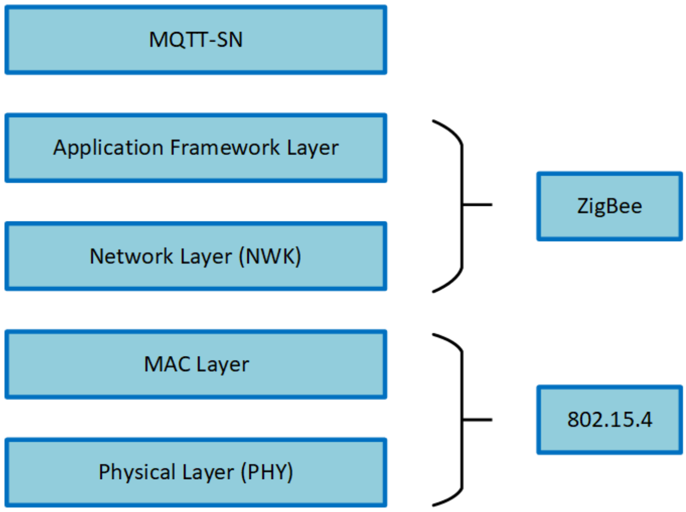

Figure 1.4: Example communication stack for an IoT smart irrigation system. Source: https://arxiv.org/abs/2009.12747.

1.7 Actuators and irrigation control

An irrigation decision becomes operational only when it is translated into a physical action. Common actuators include pumps, solenoid valves, relays, fertigation injectors, sprinklers, drip lines, and motorized valves. Actuation can be manual, semi-automatic, or fully automatic. In all cases, safety is important because wrong commands can waste water, flood the root zone, damage equipment, or create electrical risk.

Control logic may be rule-based, model-based, or AI-assisted. A rule-based controller can activate irrigation when soil moisture falls below a threshold. A model-based controller can estimate crop water requirement using evapotranspiration and water balance. An AI-assisted controller can learn patterns from historical data. The more autonomous the system becomes, the more important it is to include validation, fault detection, override mechanisms, and logging.

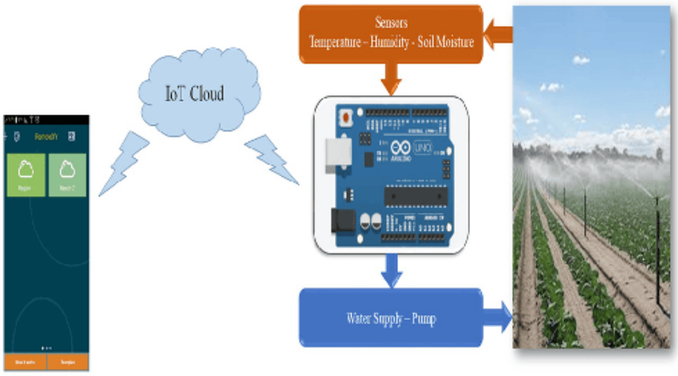

Figure 1.5: IoT-based smart irrigation system design including sensors, controller, and water pump. Source: https://arxiv.org/abs/2005.04158.

1.8 Challenges of smart irrigation systems

The literature identifies several barriers to the adoption of smart irrigation. Technical challenges include sensor calibration, power autonomy, wireless reliability, missing data, interoperability, cybersecurity, and robustness under outdoor conditions. Agronomic challenges include soil heterogeneity, crop-specific water demand, root depth variation, evapotranspiration estimation, salinity, and the difficulty of validating decisions across seasons.

Economic and social challenges are also important. Sensors and communication infrastructure may be costly for small farms. Farmers may reject systems that are difficult to maintain or that provide recommendations without clear explanation. Therefore, smart irrigation research must consider usability, trust, interpretability, maintenance, and cost, not only algorithmic performance.

1.9 Conclusion

This chapter presented the domain background of precision agriculture and smart irrigation. Precision agriculture was introduced as a management strategy based on variability and decision support. Smart irrigation was then described as the application of sensing, communication, data processing, and control to water management. The chapter also summarized key variables, communication technologies, actuators, and challenges. These concepts prepare the next chapter, which introduces Digital Twins, IoT, and artificial intelligence as enabling technologies for data-driven agricultural systems.

Chapter 2

Digital Twins, IoT, and Artificial Intelligence

2.1 Introduction

Digital agriculture uses connected devices, data models, simulation, and artificial intelligence to improve observation and decision-making in agricultural systems. After introducing precision agriculture and smart irrigation, this chapter presents the technological concepts that support this field: Digital Twins, the Internet of Things, and artificial intelligence. The goal is to define these concepts clearly before moving to the state of the art.

Digital Twin research is important because agriculture is dynamic and uncertain. Soil water content changes with irrigation, rainfall, evaporation, drainage, root uptake, and soil properties. Sensors can describe part of this reality, but decisions often require a virtual representation that can organize data, simulate scenarios, and support prediction. This is why the Digital Twin concept is increasingly studied in smart farming and water management.

Artificial intelligence is introduced here in a broad sense: not only machine learning models for prediction, but also deep learning, large language models, and multi-agent systems. These families of methods have different roles. Machine learning is useful for tabular prediction and classification. Deep learning can learn representations from images, time series, and high-dimensional data. Large language models can support natural-language interaction and knowledge access. Multi-agent systems can model distributed decision-making and coordination.

2.2 Definition of Digital Twin

The Digital Twin concept is used in many domains, but the literature generally agrees on three elements: a physical object or process, a digital representation, and a connection that keeps the digital representation updated using data. Fuller et al. [4] describe Digital Twins as digital replicas connected to physical systems through data and communication technologies. In manufacturing, Kritzinger et al.

[5] emphasize the level of data integration between the physical and digital objects.

For a background chapter, the important point is that a Digital Twin is more than a static model. A static simulation can represent a field or irrigation process, but it is not necessarily synchronized with the real system. A Digital Twin requires a data connection that updates the digital representation, and advanced forms can also generate recommendations or actions that influence the physical system.

Table 2.1: Definitions and interpretation of Digital Twin concepts

Concept	Literature meaning	Interpretation for agriculture
Physical entity	Real asset, process, or environment being represented	Soil, crop, irrigation network, greenhouse, farm, or water system
Digital representation	Model, data structure, or simulation of the physical entity	Virtual field state, water-balance model, crop model, or dashboard state
Data connection	Data flow linking physical and digital sides	Sensor readings, weather data, remote sensing, actuator feedback
Synchronization	Updating the digital side when the physical state changes	Soil moisture and weather updates modify the virtual state
Feedback	Recommendation or command from digital side toward operation	Decision support, warning, irrigation schedule, or actuator instruction

2.3 Digital Model, Digital Shadow, and Digital Twin

Kritzinger et al. [5] propose a useful distinction between Digital Model, Digital Shadow, and Digital Twin. A Digital Model has no automatic data exchange with the physical object. A Digital Shadow has automatic data flow from the physical object to the digital object. A Digital Twin has automatic bidirectional flow, so that digital and physical sides can influence each other.

This distinction is important because many systems called Digital Twins are actually monitoring dashboards or decision support tools with limited synchronization. In agriculture, a digital crop model that uses manually entered data is closer to a Digital Model. A sensor dashboard that updates soil moisture automatically is closer to a Digital Shadow. A synchronized system that uses data to simulate, recommend, and potentially control irrigation approaches the Digital Twin level.

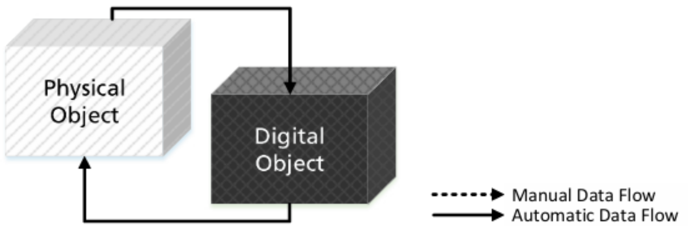

Figure 2.1: Digital Model, Digital Shadow, and Digital Twin integration levels. Source: https://doi.org/10.1016/j.ifacol.2018.08.474.

Table 2.2: Data-flow distinction between digital representations

Type	Physical-to-digital flow	Digital-to-physical flow	Typical example
Digital Model	Manual or offline	Manual or absent	Standalone crop simulation
Digital Shadow	Automatic	Manual or absent	Sensor dashboard updated from field data
Digital Twin	Automatic	Automatic or operationally connected	Synchronized decision/control system

2.4 Digital Twin architecture layers

A Digital Twin architecture is usually layered. The physical layer contains the real process or asset. The sensing layer collects measurements. The communication layer transfers data. The data layer stores and cleans information. The modeling layer represents the system state. The simulation layer evaluates scenarios. The analytics or AI layer supports prediction and optimization. The application layer presents results to users. Finally, the feedback or control layer closes the loop through recommendations or actuator commands.

In agriculture, these layers must handle biological and environmental variability. Unlike a factory machine, a crop is a living system affected by weather, soil, pests, growth stage, and management practices. Therefore, agricultural Digital Twins require both engineering infrastructure and agronomic knowledge.

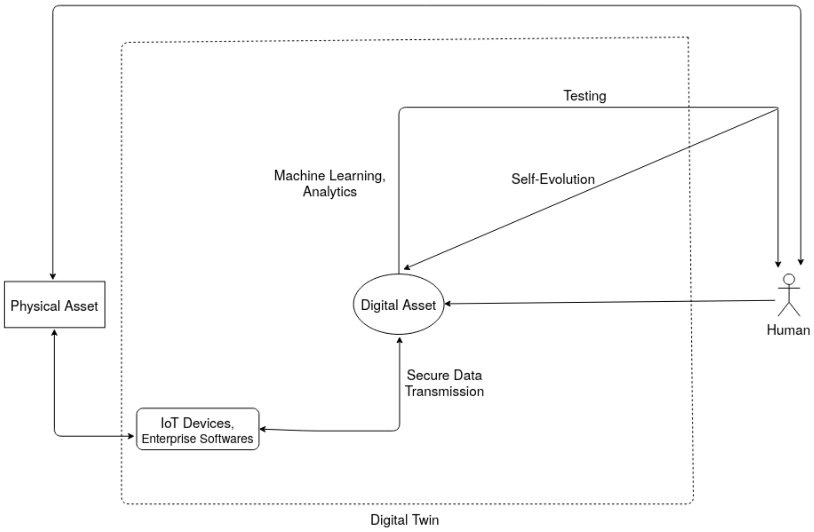

Figure 2.2: Digital Twin reference framework with components and information flows. Source: https://arxiv.org/abs/2011.02833.

2.5 Internet of Things as the physical-to-digital bridge

The Internet of Things (IoT) provides the physical-to-digital connection required by smart irrigation and Digital Twins. Gubbi et al. [6] describe IoT as a networked environment where sensing and actuating devices blend with the physical world and share information across platforms. In agriculture, IoT devices can include soil sensors, weather stations, cameras, embedded controllers, gateways, and actuator modules.

IoT is important because a Digital Twin cannot remain synchronized without data. Sensors measure soil and environmental conditions, gateways aggregate and transmit messages, and cloud or edge services store and process them. In irrigation, IoT also supports feedback because commands can be sent to pumps and valves, while status messages can confirm execution.

Table 2.3: IoT functions in Digital Twin and irrigation systems

Function	Description	Example in agriculture
Sensing	Measuring physical variables	Soil moisture, air temperature, rainfall
Identification	Associating data with device, zone, or asset	Sensor ID, plot ID, timestamp
Communication	Sending data between devices and services	LoRa, Wi-Fi, MQTT, cellular
Edge processing	Cleaning or aggregating data near the field	Gateway validation, local filtering
Cloud/data service	Storing and exposing data for applications	Database, API, dashboard backend
Actuation	Applying commands to physical devices	Pump, valve, relay, fertigation unit

2.6 Artificial intelligence: general definition

Artificial intelligence is a broad field concerned with building systems that perform tasks associated with perception, reasoning, learning, communication, planning, and action. The Dartmouth proposal that introduced the field framed AI around the idea that aspects of learning and intelligence could be described precisely enough for machines to simulate them [7]. Modern AI includes symbolic reasoning, search, probabilistic models, machine learning, deep learning, language models, and agent systems.

In agricultural decision support, AI is useful when the decision cannot be expressed only as a simple fixed rule. Irrigation depends on nonlinear relationships between weather, soil, crop, and management actions. AI methods can learn from historical data, detect patterns, classify risk, predict future states, or help users interact with complex data through natural language.

2.7 Machine learning

Machine learning (ML) is a subfield of AI in which models improve their performance by learning patterns from data. In irrigation, ML can be used for classification, regression, forecasting, anomaly detection, and optimization. Examples include predicting whether irrigation is needed, estimating future soil moisture, forecasting evapotranspiration, detecting sensor faults, or estimating irrigation duration.

Classical ML methods such as Decision Trees, Random Forests, Support Vector Machines, Gradient Boosting, and k-nearest neighbors are often suitable for tabular sensor data. Random Forests are especially common because they combine many decision trees and can model nonlinear relationships while remaining easier to train than many deep learning architectures [8].

Table 2.4: Machine learning tasks in irrigation and agriculture

Task	Typical model family	Example output
Classification	Decision Tree, Random Forest, SVM	Irrigate / do not irrigate
Regression	Random Forest, Gradient Boosting, neural network	Predicted soil moisture or water amount
Time-series forecasting	ARIMA, LSTM, temporal neural network	Future moisture or evapotranspiration
Anomaly detection	Isolation Forest, statistical threshold	Sensor fault or abnormal reading
Optimization	Reinforcement learning, evolutionary algorithms	Irrigation policy or schedule

2.8 Deep learning

Deep learning (DL) is a machine learning approach based on neural networks with multiple processing layers. LeCun, Bengio, and Hinton [9] explain that deep learning learns representations at multiple levels of abstraction. This makes it powerful for unstructured or high-dimensional data such as images, audio, text, and long time series.

In agriculture, deep learning is widely used for crop disease detection, weed recognition, fruit detection, yield estimation, remote sensing image analysis, and time-series prediction. In irrigation, recurrent networks, temporal convolutional networks, and LSTM models can be used to forecast soil moisture or water demand. However, deep learning usually requires larger datasets, more computation, and stronger validation than classical ML.

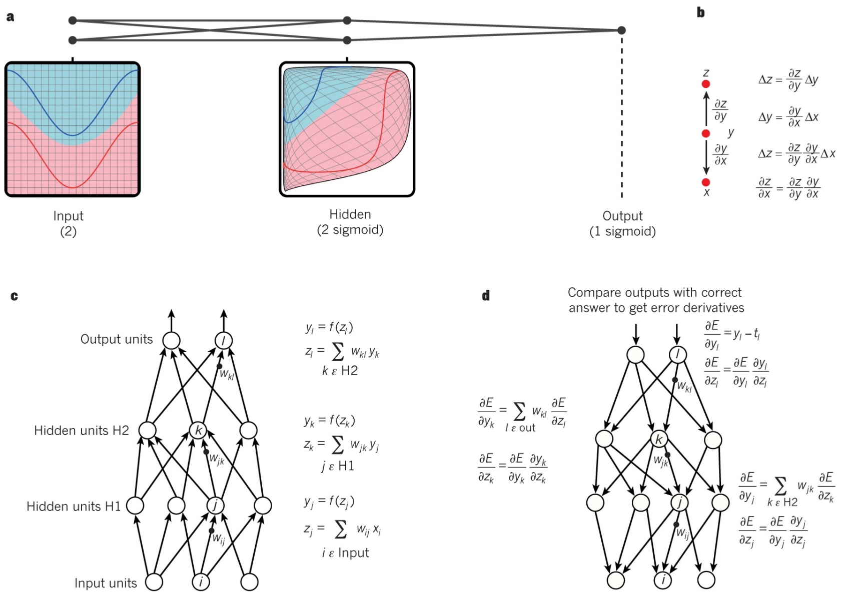

Figure 2.3: Multilayer neural networks and backpropagation. Source: https://www.nature.com/articles/nature14539.

2.9 Large language models

Large language models (LLMs) are deep learning models trained on large text corpora to predict, generate, summarize, translate, and reason over language. They are part of the broader family of foundation models, which Bommasani et al. [10] describe as models trained at scale and adaptable to many downstream tasks. Modern LLMs are usually based on the Transformer architecture introduced by Vaswani et al. [11]. Brown et al. [12] showed that scaling autoregressive language models can improve few-shot behavior, meaning that tasks can be specified through instructions and examples in text.

LLMs are not irrigation models by themselves. Their value in agricultural systems is mainly at the interaction and knowledge layer. They can help users query sensor histories, summarize alerts, explain model outputs, generate reports, retrieve agronomic documentation, or support natural-language interfaces. They must be used carefully because generated text can be incorrect, unsupported, or overconfident. For technical and agronomic decisions, LLM outputs should be grounded in validated data, citations, and human review.

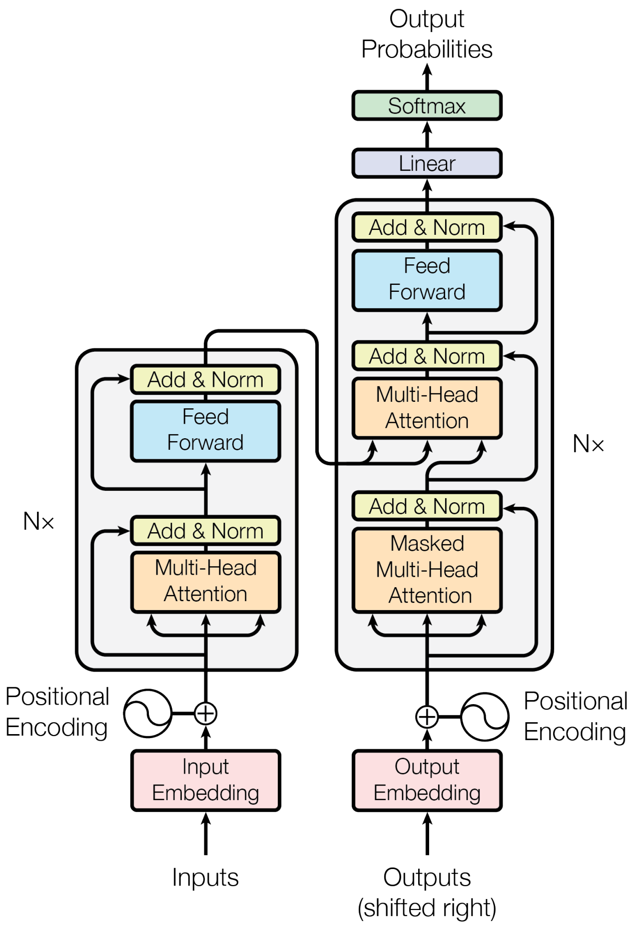

Figure 2.4: The Transformer model architecture. Source: https://arxiv.org/abs/1706.03762.

2.10 Multi-agent systems

Multi-agent systems (MAS) study systems composed of multiple interacting agents. Wooldridge and Jennings [13] describe intelligent agents through properties such as autonomy, reactivity, proactiveness, and social ability. In a multi-agent system, several agents may observe, communicate, negotiate, coordinate, or act in a shared environment.

In agriculture, a multi-agent view can be useful when decisions are distributed. For example, one agent may monitor weather, another may evaluate soil state, another may optimize irrigation scheduling, and another may communicate with a human operator. In collective irrigation networks, different farms or zones may compete for limited water, making coordination and negotiation important.

Recent work also connects LLMs with agent systems. Foundation models can provide language understanding and planning capabilities, while agent frameworks organize tool use, memory, roles, and communication. However, the classical MAS literature remains important because it provides established concepts for coordination, autonomy, communication protocols, and decentralized decision-making.

Table 2.5: AI families and their possible roles in agricultural decision support

AI family	Main idea	Possible role in agriculture
Machine learning	Learn prediction rules from structured data	Irrigation classification, soil moisture prediction
Deep learning	Learn representations from high-dimensional data	Image analysis, disease detection, time-series forecasting
Large language models	Generate and reason over natural language	Explanations, reporting, question answering, user interface
Multi-agent systems	Coordinate autonomous or semi-autonomous agents	Distributed monitoring, scheduling, negotiation, control

2.11 Human-in-the-loop decision support

Agricultural AI and Digital Twins should not be treated as replacements for agronomic expertise. Farmers and agronomists know local practices, crop history, irrigation constraints, and risk tolerance. Human-in-the-loop decision support keeps the human user involved in interpretation, validation, and action, especially when physical actuators are involved.

This approach is particularly important for irrigation because wrong recommendations can have physical and economic consequences. A useful system should therefore provide evidence, confidence, explanations, historical context, and manual override options. Automation can increase efficiency, but trust depends on transparency, validation, and the ability to correct or reject recommendations.

2.12 Challenges

Digital Twin and AI adoption in agriculture faces several challenges. Data may be noisy, missing, poorly calibrated, or collected at different time scales. Models trained in one crop, soil type, or climate may not generalize to another. Communication networks may be unreliable in rural areas. Security and privacy are also important because connected systems can expose farm data or actuator control channels.

AI introduces additional risks: overfitting, lack of interpretability, bias in datasets, weak field validation, and excessive confidence in predictions. LLMs add the risk of unsupported generation. Multi-agent systems add coordination and safety challenges. For these reasons, background concepts must be connected to validation methods in later chapters.

2.13 Conclusion

This chapter defined Digital Twins, IoT, and artificial intelligence as enabling concepts for data-driven agricultural systems. Digital Twins provide synchronized virtual representations of physical processes. IoT provides the sensing, communication, and actuation bridge. AI provides methods for prediction, representation learning, language interaction, and distributed coordination. These foundations support the next part of the thesis, where the literature is reviewed and compared.

Part III

State of the Art

Chapter 3

Digital Twins in Industry

This part reviews concrete previous works that are close to the technological foundations of a smart irrigation digital twin system. It begins with industrial digital twins, because Industry 4.0 provides many mature examples of synchronization between a physical asset and a virtual model. It then moves to smart agriculture and irrigation systems, before focusing on digital twins applied to agriculture and water management. The selected works are implementation, prototype, simulation, or case-study papers. Review and survey papers are not used as main state-of-the-art papers in this part.

3.1 Cyber-Physical Production Systems

3.1.1 The Digital Twin: Realizing the Cyber-Physical Production System for Industry 4.0 [14]

Overview

Uhlemann et al. address the problem of connecting a physical production system with a digital representation in the context of Industry 4.0. The paper is relevant because it treats the digital twin as a practical mechanism for realizing a cyber-physical production system, rather than as a purely conceptual model. The physical asset is an industrial production environment, while the digital counterpart is used to represent and analyze production behavior.

Architecture / Methodology

The methodology is organized around the link between a physical production system and its virtual counterpart. The physical side contains the machines, production resources, and shop-floor events. The digital side receives data from these resources and uses them to keep an updated representation of the production state.

The important point in this paper is the data flow. Measurements and events are not treated as isolated monitoring values. They are used to update the virtual model so that the model reflects what is happening in the production system.

This is the difference between a simple dashboard and a digital twin: the virtual representation must be continuously informed by the physical process.

The paper also shows that the digital twin has a decision-oriented role. Once the virtual model is synchronized with the physical system, it can be used to understand production behavior, detect deviations, and support operational decisions. For this thesis, the same logic can be translated to irrigation: field sensors feed a virtual state, and the virtual state supports water-management decisions.

Evaluation and Results

The paper presents a concrete cyber-physical production-system realization and uses it to illustrate how digital twin data can support monitoring and production understanding. The available verified metadata does not expose detailed numerical performance metrics such as latency, prediction accuracy, or throughput improvement. Therefore, in this review the validation is treated as qualitative prototype validation.

Limitations and Relevance to This Project

The main limitation is that the paper is focused on industrial production and does not address uncertain biological processes such as soil moisture dynamics, crop water demand, or weather variability. However, it is useful because it clarifies the core digital twin pattern: a physical asset must continuously inform a virtual representation, and the virtual representation must support decisions. The remaining gap for irrigation is to adapt this pattern to field sensing, water management, and agronomic decision support.

3.2 Visualization and Human Interaction

3.2.1 Visualising the Digital Twin Using Web Services and Augmented Reality [15]

Overview

Schroeder et al. study how a digital twin can be visualized and accessed by users through web services and augmented reality. The paper is important because digital twins are not only data models; they also need interfaces that allow operators to understand the state of the physical system. In industrial monitoring, this interaction layer helps users interpret data and inspect the virtual representation of equipment.

Architecture / Methodology

The methodology separates the digital twin system into several clear layers. The first layer is the physical industrial asset, which produces operational data. The second layer is the digital representation, where the state of the asset is modeled and made available to external services. The third layer is the web-service interface, which exposes the digital twin data in a structured way.

The final layer is the augmented-reality visualization. Instead of showing only numerical values, the system allows the user to inspect information through a visual interface connected to the asset. This makes the digital twin easier to understand for an operator, especially when the physical object and its digital information must be interpreted together.

This paper is useful because it treats visualization as part of the digital twin methodology. A virtual model is not sufficient if users cannot interpret it. In a smart irrigation system, the same principle applies: soil moisture values, irrigation state, and recommendations must be displayed in a way that helps the user understand the field situation.

Evaluation and Results

The work is evaluated as a prototype that demonstrates the feasibility of visualizing a digital twin through web services and augmented reality. The paper validates the interaction concept qualitatively. The available verified metadata does not provide exact numerical values for response time, usability scores, or industrial productivity gains.

Limitations and Relevance to This Project

The limitation is that the contribution is mainly centered on visualization and interaction, not on predictive modeling or automatic control. For a smart irrigation digital twin, this work remains relevant because farmers and operators also need clear interfaces to understand sensor data, virtual states, and recommendations. The gap is the integration of visualization with irrigation-specific prediction, scheduling, and control.

3.3 Production Management and Control

3.3.1 Digital Twin-Based Smart Production Management and Control Framework for the Complex Product Assembly Shop-Floor [16]

Overview

Zhuang et al. propose a digital twin framework for smart production management and control in a complex product assembly shop floor. The work is relevant because it goes beyond simple visualization and uses the digital twin as a management and control support layer. The physical asset is an assembly shop floor, and the virtual model represents the state of the assembly process.

Architecture / Methodology

The paper identifies four main methodological blocks. The first block is the real-time acquisition, organization, and management of physical shop-floor data. This block is responsible for collecting data from the assembly environment and preparing it so that it can be used by the digital twin.

The second block is the construction of the assembly shop-floor digital twin. This virtual representation models the resources, state, and behavior of the assembly process. It is not only a static model of the factory layout; it is designed to reflect the current production situation.

The third block is prediction based on the digital twin and big-data methods. The digital twin provides structured production-state information, while the prediction layer uses this information to anticipate production behavior. The fourth block is the management and control service layer, which transforms analysis results into operational support for the shop floor.

The implementation scenario is a satellite assembly shop floor. This example shows how a digital twin can support a complex process where many resources, operations, and constraints must be coordinated. For irrigation, the equivalent methodological chain is sensor acquisition, field-state modeling, water-need prediction, and irrigation decision support.

Evaluation and Results

The paper validates the framework through an assembly shop-floor scenario. The evaluation demonstrates that digital twin modeling can support production management and control by connecting real production data with predictive and service functions. The verified public information does not provide detailed numerical performance values, so this review treats the reported validation as scenario-based and qualitative.

Limitations and Relevance to This Project

The limitation is that the framework is designed for manufacturing operations where the production process is more controlled than an agricultural field. It does not address weather uncertainty, soil heterogeneity, or plant response. Its relevance is strong at the architectural level: a smart irrigation digital twin also needs data acquisition, virtual modeling, prediction, and decision services. The remaining gap is to translate this layered industrial logic into a practical irrigation workflow.

3.4 Open-Source Smart Manufacturing

3.4.1 An Open Source Approach to the Design and Implementation of Digital Twins for Smart Manufacturing [17]

Overview

Damjanovic-Behrendt and Behrendt propose an open-source approach for designing and implementing digital twins in smart manufacturing. This paper is relevant because it focuses on implementation choices, not only on abstract digital twin definitions. In industrial systems, open-source components can reduce vendor lock-in and make the digital twin architecture easier to reproduce.

Architecture / Methodology

The methodology is implementation-oriented. The authors focus on how a digital twin can be built with open-source components rather than only describing a conceptual architecture. This makes the paper useful for projects that need a practical and reproducible system design.

The physical manufacturing resources provide the operational data. These data are connected to digital representations through software services. The digital twin layer then organizes the information so that it can be accessed by applications that support monitoring, analysis, and decision-making.

The paper also emphasizes interoperability. In smart manufacturing, different machines, protocols, and software services must communicate. The open-source approach is therefore not only a cost decision; it is a way to make the system easier to integrate, extend, and reproduce.

For a smart irrigation digital twin, this methodology suggests that the architecture should be modular. Sensors, backend services, database components, prediction modules, and user interfaces should communicate through clear interfaces so that the system can evolve without being rebuilt from zero.

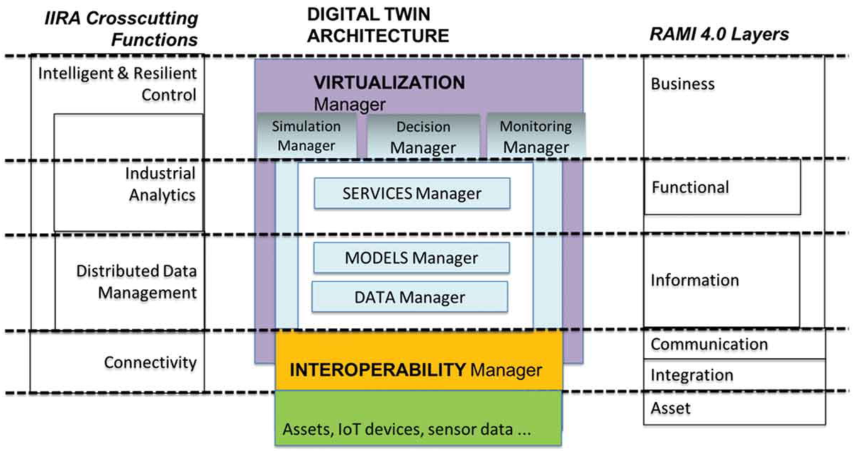

Figure 3.1: Open-source digital twin implementation approach for smart manufacturing, adapted from [17]. Source: https://doi.org/10.1080/0951192X.2019.1599436.

Evaluation and Results

The paper provides an implementation case for smart manufacturing. The validation is mainly architectural and prototype-oriented. The available verified metadata does not expose exact numerical metrics such as prediction accuracy, downtime reduction, or throughput improvement, so the result is considered qualitative validation of an open-source implementation path.

Limitations and Relevance to This Project

The limitation is that the work targets manufacturing systems, not agriculture. It also does not focus on irrigation decisions, agronomic constraints, or water-saving metrics. Its relevance lies in the engineering approach: open-source components, modular services, and reproducible implementation are useful for building practical digital twin systems outside industry. The gap remains the adaptation of these implementation ideas to IoT sensing, irrigation scheduling, and water management.

3.5 Discussion

The four industrial papers show a progression from synchronization between a physical production system and a virtual representation, to visualization, prediction, management, and implementation. They are useful because they demonstrate that a digital twin is not a single algorithm. It is a system that connects data acquisition, virtual modeling, decision support, and sometimes control. However, these works are situated in industrial environments where assets and processes are more stable than agricultural systems. Agriculture adds uncertainty from weather, soil, crop growth, and water availability.

Table 3.1: Comparison of selected industrial digital twin papers

Paper	Industrial domain	Digital twin purpose	Data source	Modeling / AI method	Validation	Main result	Limitation
Uhlemann et al. [14]	Cyber-physical production	Realize Industry 4.0 production-system twin	Shop-floor production data	Virtual production-system representation	Prototype demonstration	Shows how CPPS data can feed a digital counterpart	No agricultural or water-management context
Schroeder et al. [15]	Industrial monitoring	Visualize digital twin state	Web-service data from physical asset	Service-based representation and AR visualization	Prototype demonstration	Provides an interaction layer for digital twin inspection	Limited focus on prediction and control
Zhuang et al. [16]	Complex assembly shop floor	Production management and control	Real-time shop-floor data	Digital twin plus big-data-driven prediction	Assembly scenario	Connects data acquisition, twin model, prediction, and control services	Manufacturing assumptions do not transfer directly to fields
Damjanovic-Behrendt and Behrendt [17]	Smart manufacturing	Open-source digital twin implementation	Industrial system data	Modular open-source implementation	Implementation case	Supports reproducible smart-manufacturing twin design	Does not target irrigation or agronomic decisions

The main gap identified from the industrial literature is the need to adapt mature digital twin principles to agricultural constraints. Industrial digital twins show strong patterns for synchronization, service layers, and decision support, but they do not solve the specific problems of irrigation: sparse field sensing, environmental variability, water balance, and user-oriented irrigation control.

Chapter 4

Smart Agriculture and Irrigation

This chapter reviews concrete smart agriculture and irrigation systems that use sensors, communication networks, controllers, or machine learning to support irrigation decisions. These works are important because they establish the sensing and automation layer required before a complete irrigation digital twin can be built.

4.1 Wireless Sensor Network Irrigation

4.1.1 Remote Sensing and Control of an Irrigation System Using a Distributed Wireless Sensor Network [18]

Overview

Kim et al. propose a distributed wireless sensor network for remote sensing and control of an irrigation system. The work addresses the difficulty of monitoring field conditions and controlling irrigation equipment without continuous manual inspection. It is relevant because it connects field sensing with irrigation actuation, which is one of the basic requirements of smart irrigation.

Architecture / Methodology

The system is organized around distributed sensor nodes installed in the field. These nodes are responsible for collecting local measurements that describe the irrigation environment. The purpose is to reduce the need for manual inspection and to provide the controller with field information at different locations.

The communication layer connects the distributed nodes to the irrigation control unit. This layer is important because field sensors are useful only if their measurements can be transmitted reliably to the place where irrigation decisions are made. The paper therefore treats wireless communication as a central part of the irrigation architecture.

The control component uses the received data to support remote irrigation operation. In this type of system, the methodological chain is simple but important: collect field data, transmit the data, interpret the measurements, and control the irrigation equipment. This chain is a foundation for later digital twin systems, where the same data would also update a virtual field state.

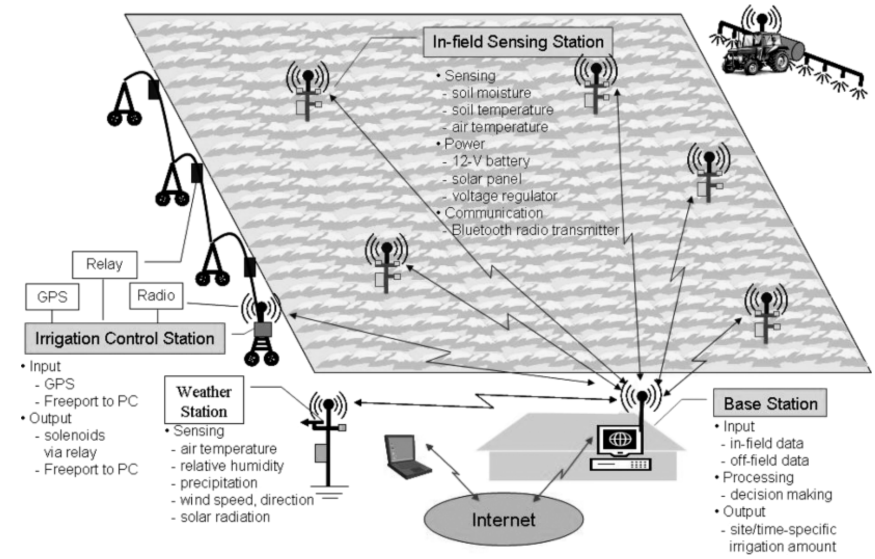

Figure 4.1: Distributed wireless sensor network for remote irrigation sensing and control, adapted from [18]. Source: https://doi.org/10.1109/TIM.2008.917198.

Evaluation and Results

The paper evaluates the system as a practical wireless sensor network and control prototype for irrigation. It demonstrates the feasibility of using distributed sensing for remote irrigation monitoring and control. The verified metadata does not provide exact values for water saving, packet delivery, or crop yield improvement, so this review treats the validation as prototype-based and qualitative.

Limitations and Relevance to This Project

The main limitation is that the work is mainly an irrigation monitoring and control system, not a synchronized digital twin with a virtual model and predictive simulation. It is useful because it shows how field sensors and control equipment can be connected in an irrigation context. The remaining gap is to combine this sensing-control loop with virtual representation, prediction, and decision support.

4.1.2 Automated Irrigation System Using a Wireless Sensor Network and GPRS Module [19]

Overview

Gutierrez et al. present an automated irrigation system that combines a wireless sensor network with a GPRS communication module. The paper addresses the need for remote operation in agricultural fields where wired communication is difficult or expensive. Its relevance lies in the integration of field sensing, long-range communication, and automatic irrigation control.

Architecture / Methodology

The architecture begins with field sensor nodes that collect measurements from the agricultural environment. These nodes form a wireless sensor network, which allows the system to cover a field without wired infrastructure. This is important for agricultural deployment because farms are often spatially distributed and difficult to cable.

The GPRS module extends the communication range. Instead of keeping the system limited to local communication, GPRS allows data to be sent to remote users or services. This makes the irrigation system more suitable for remote supervision and control.

The irrigation actuation layer closes the loop. Measurements collected in the field are transmitted through the network, processed by the control unit, and used to activate or deactivate irrigation. The methodology therefore combines sensing, communication, decision, and actuation in one automated irrigation workflow.

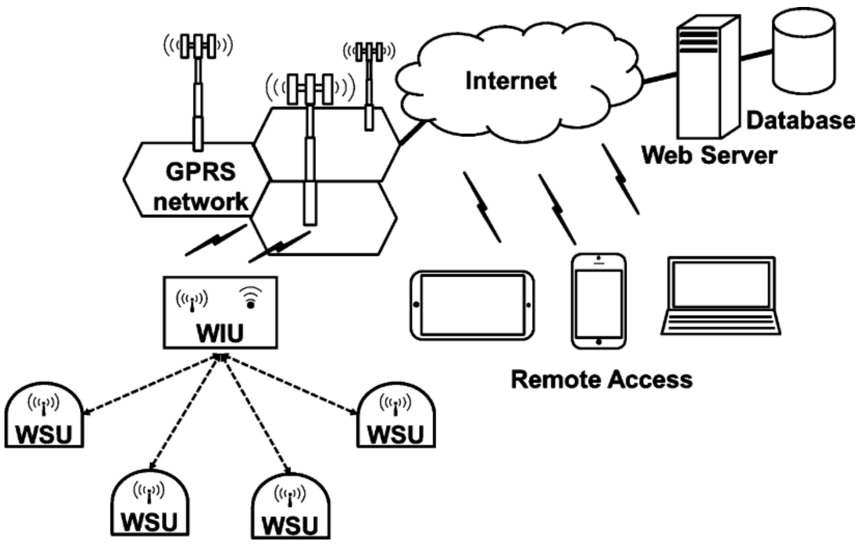

Figure 4.2: Automated irrigation architecture using WSN and GPRS communication, adapted from [19]. Source: https://doi.org/10.1109/TIM.2013.2276487.

Evaluation and Results

The work is evaluated as an automated irrigation prototype. It validates that WSN and GPRS technologies can be used to support remote irrigation management. The verified metadata does not expose exact numerical results for water savings, communication reliability, or energy consumption. Therefore, the result is considered qualitative validation of the architecture.

Limitations and Relevance to This Project

The limitation is that the system focuses on automation and remote communication, but it does not include a full digital twin layer or advanced prediction. It remains relevant because it demonstrates a practical field-communication pattern for irrigation systems. The gap is to enrich this pattern with real-time virtual state, irrigation simulation, and intelligent recommendation.

4.2 Low-Cost IoT Irrigation

4.2.1 A Low Cost Smart Irrigation System Using MQTT Protocol [20]

Overview

Kodali and Sarjerao propose a low-cost smart irrigation system using the MQTT protocol. The paper addresses the need for affordable irrigation automation based on lightweight IoT communication. It is relevant because MQTT is widely used in IoT systems where small devices need to publish sensor readings and receive commands.

Architecture / Methodology

The methodology is based on a low-cost IoT architecture. A field device collects irrigation-related measurements and communicates them to the application layer. The goal is to make the system affordable while still supporting automatic irrigation decisions.

MQTT is the central communication protocol. It uses a publish-subscribe model, where field devices publish sensor readings and application components subscribe to the topics they need. The same model can also be used to send control commands back to the device. This is useful in irrigation because sensor messages are usually small and frequent.

The control logic connects communication with actuation. Once the application receives the sensor data, it can decide whether the pump or valve should be activated. The paper therefore demonstrates how a lightweight messaging protocol can support a complete sensing-to-control irrigation loop.

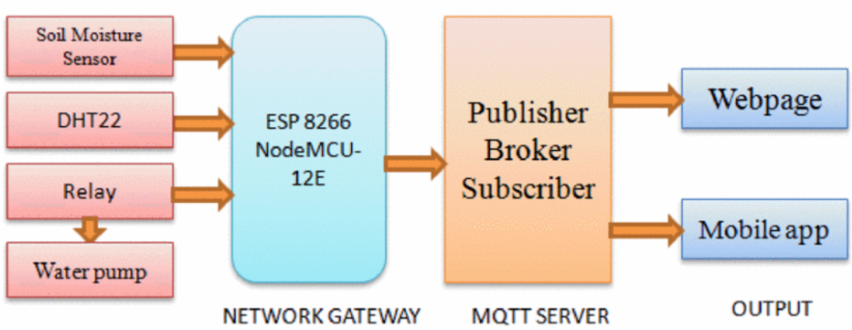

Figure 4.3: Low-cost MQTT-based smart irrigation architecture, adapted from [20]. Source: https://doi.org/10.1109/TENCONSpring.2017.8070095.

Evaluation and Results

The paper validates the system as a low-cost prototype. It demonstrates that MQTT can support basic irrigation monitoring and control. The verified metadata does not provide detailed numeric measurements for latency, water saving, or field-scale performance, so the validation is treated as qualitative prototype validation.

Limitations and Relevance to This Project

The limitation is that the system is mainly an IoT control prototype. It does not include digital twin synchronization, water-balance simulation, or machine-learning decision support. Its relevance is high at the communication level because MQTT is appropriate for connecting field devices to a backend. The gap is to build a higher-level decision and virtual representation layer on top of the IoT communication.

4.3 Machine Learning for Irrigation Decision Support

4.3.1 An IoT Based Smart Irrigation Management System Using Machine Learning and Open Source Technologies [21]

Overview

Goap et al. propose an IoT-based smart irrigation management system that uses machine learning and open-source technologies. The problem addressed is the improvement of irrigation decisions by combining sensed data with learning-based prediction. This work is relevant because it moves beyond rule-based irrigation and introduces data-driven decision support.

Architecture / Methodology

The methodology combines IoT sensing with machine-learning decision support. The IoT layer collects data from the agricultural environment. These data may include soil and environmental variables that influence crop water needs. The data are then sent to a processing layer where they can be cleaned, organized, and prepared for decision-making.

The machine-learning layer is the main difference between this work and simpler rule-based irrigation systems. Instead of relying only on fixed thresholds, the system uses learned relationships between input variables and irrigation decisions. This allows the decision process to consider several variables together.

The application layer presents or applies the irrigation decision. In this workflow, the system moves from sensing to data processing, then to prediction or classification, and finally to irrigation management. The use of open-source technologies makes the architecture easier to reproduce and adapt.

For a digital twin project, this methodology is important because machine learning can become part of the virtual decision layer. However, a digital twin still needs a synchronized virtual representation in addition to prediction.

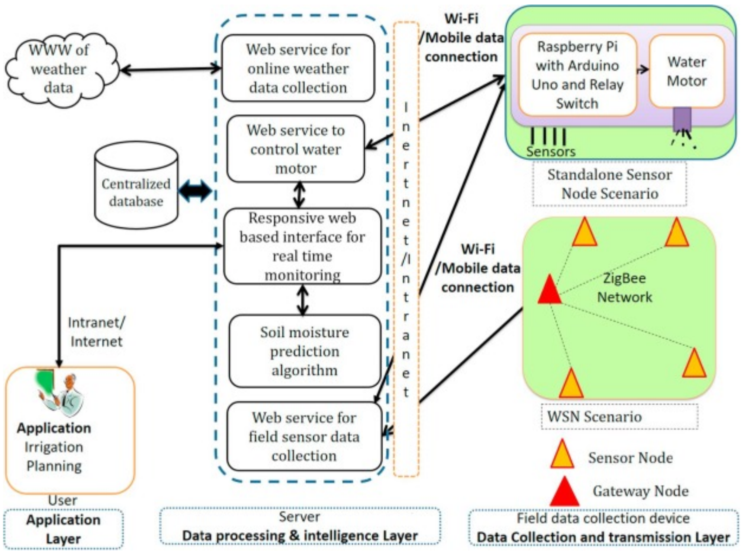

Figure 4.4: IoT and machine-learning workflow for smart irrigation management, adapted from [21]. Source: https://doi.org/10.1016/j.compag.2018.09.040.

Evaluation and Results

The paper evaluates the proposed smart irrigation management system as an IoT and machine-learning implementation. It demonstrates that machine learning can be integrated with open-source IoT components for irrigation decision support. The verified metadata does not provide exact performance values for prediction accuracy, water saving, or yield improvement. Therefore, this review describes the result as implementation validation rather than a confirmed field-scale performance benchmark.

Limitations and Relevance to This Project

The limitation is that the paper focuses on smart irrigation decision support, not on a complete digital twin with continuous physical-virtual synchronization. It is useful because it shows how machine learning can be placed between sensor data and irrigation decisions. The remaining gap is to connect this decision layer to a virtual representation of the field and to a real-time monitoring and control workflow.

4.4 Discussion

The selected smart irrigation papers show that soil and environmental sensing, wireless communication, and automation are already practical in irrigation systems. Wireless sensor networks and GPRS make field monitoring possible over distributed areas. MQTT provides a lightweight IoT communication model. Machine learning adds the possibility of data-driven irrigation decisions. However, these systems usually remain at the level of monitoring, control, or decision support. They do not fully model a synchronized virtual copy of the irrigation system, and they rarely combine sensing, prediction, simulation, user decision support, and actuation into one complete digital twin.

Table 4.1: Comparison of selected smart irrigation papers

Paper	Application	Sensors / data	Communication	Irrigation decision method	Validation	Main result	Limitation
Kim et al. [18]	Remote irrigation sensing and control	Distributed field measurements	Wireless sensor network	Control based on sensed field data	Prototype	Shows feasibility of remote sensing and control	No digital twin or predictive simulation
Gutierrez et al. [19]	Automated irrigation	Field sensor-node data	WSN and GPRS	Automated controller using sensor data	Prototype	Demonstrates remote automated irrigation architecture	Limited virtual modeling and AI support
Kodali and Sarjerao [20]	Low-cost smart irrigation	IoT sensor readings	MQTT	Lightweight publish-subscribe control logic	Prototype	Shows MQTT suitability for low-cost irrigation IoT	Basic control without digital twin layer
Goap et al. [21]	Smart irrigation management	IoT and environmental data	IoT network	Machine-learning decision support	Implementation	Integrates ML with open-source IoT irrigation	No full physical-virtual synchronization

The synthesis of gaps is clear. Smart irrigation systems already handle sensing, connectivity, and automation, but they often lack a persistent virtual model of the field state. They also tend to treat irrigation as a direct control problem rather than as a digital twin problem where the physical field, virtual representation, prediction layer, and decision workflow are synchronized.

Chapter 5

Digital Twins in Agriculture and Irrigation

This chapter reviews concrete digital twin applications in agriculture and irrigation. The selected papers are the closest previous works to a smart irrigation digital twin system because they combine agricultural sensing, virtual representation, simulation, machine learning, irrigation scheduling, or water distribution management.

5.1 Digital Twins for Irrigation Scheduling

5.1.1 IoT-Digital Twin-Inspired Smart Irrigation Approach for Optimal Water Utilization [22]

Overview

Manocha et al. propose an IoT-digital twin-inspired approach for smart irrigation and optimal water utilization. The paper addresses the problem of applying irrigation water more efficiently by combining field data with a virtual representation. It is directly relevant because it connects IoT sensing with digital twin logic in an irrigation context.

Architecture / Methodology

The proposed approach starts from the physical irrigation environment. Sensors collect information from the field and provide the data needed to understand current irrigation conditions. This physical layer is essential because the digital twin cannot remain meaningful if it is not updated from real observations.

The IoT layer transfers the collected measurements to the digital side of the system. In this step, sensor values become structured inputs for the virtual representation. The digital twin-inspired layer then uses these inputs to represent the state of the irrigation context and to reason about water needs.

The decision mechanism is oriented toward optimal water utilization. Instead of only displaying measurements, the system uses the virtual representation to support irrigation decisions. The methodology therefore follows a full chain: physical sensing, IoT communication, virtual state representation, and water-use decision support.

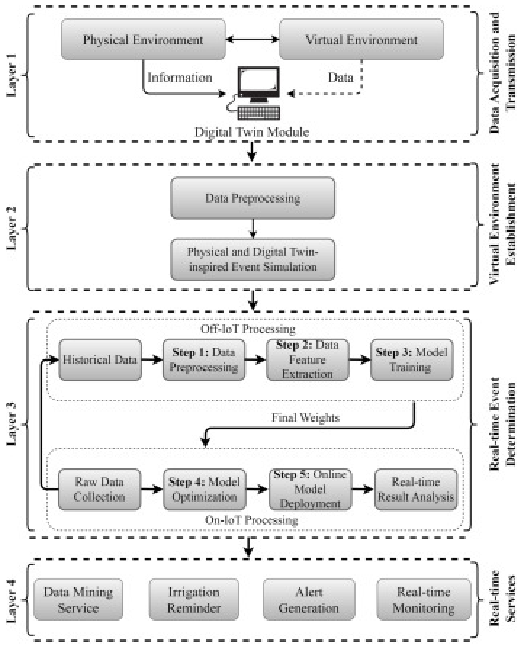

Figure 5.1: IoT-digital twin-inspired smart irrigation architecture, adapted from [22]. Source: https://doi.org/10.1016/j.suscom.2023.100947.

Evaluation and Results

The paper evaluates the approach as a concrete smart irrigation system. It reports the use of IoT and digital twin concepts for irrigation optimization. Exact numerical results are not available in the verified metadata used for this review, so the evaluation is described as system-level validation rather than a quantified benchmark.

Limitations and Relevance to This Project

The limitation is that the available public metadata does not expose detailed information about deployment scale, crop context, or exact performance metrics. Nevertheless, the paper is highly relevant because it combines IoT sensing, a virtual irrigation representation, and water utilization objectives. The remaining gap is to provide a practical implementation that clearly exposes real-time monitoring, decision support, and irrigation management in one reproducible workflow.

5.1.2 Assimilation of Sentinel-2 Biophysical Variables into a Digital Twin for the Automated Irrigation Scheduling of a Vineyard [23]

Overview

Bellvert et al. develop a digital twin for automated vineyard irrigation scheduling by assimilating Sentinel-2 biophysical variables. The problem addressed is precision irrigation under spatial variability inside a vineyard. The paper is important because it combines remote sensing, digital twin modeling, soil-water balance simulation, and irrigation scheduling.

Architecture / Methodology

The work extends an irrigation decision-support system with digital twin and IoT functions. The physical vineyard is monitored through soil and plant sensors. Meteorological information is also integrated because irrigation demand depends strongly on weather conditions.

A key methodological contribution is the assimilation of Sentinel-2 biophysical variables. The authors use satellite-derived fractional absorbed photosynthetically active radiation to update the digital representation of vineyard development. This allows the virtual system to account for spatial variability that cannot be captured by a few local sensors alone.

The simulation layer models soil-water balance components. The digital twin combines sensor data, meteorological data, and remote-sensing information to estimate water status and irrigation needs. The system then sends daily irrigation prescriptions to the irrigation controller, which makes the approach close to an operational irrigation digital twin.

This paper is methodologically strong because it shows synchronization from several data sources, simulation of the agronomic state, and connection to irrigation prescriptions. It is one of the clearest examples of a digital twin used for automated precision irrigation.

Evaluation and Results

The paper provides quantitative validation. The relationship between Sentinel-2 fAPAR and estimated fIPAR reached R^2 values between 0.61 and 0.91, with an RMSD of 0.10. The evapotranspiration comparison reported RMSD values of 0.98 mm/day and 1.14 mm/day. The study also compared automated vineyard irrigation scheduling with traditional irrigation management and reported differences in applied irrigation among sectors.

Limitations and Relevance to This Project

The limitation is that the system relies on satellite data, vineyard-specific configuration, and an established decision-support platform. This can make reproduction difficult for small low-cost deployments. Its relevance is strong because it shows how a digital twin can synchronize sensor, meteorological, and remote-sensing data and transform them into irrigation prescriptions. The gap for this project is to implement a practical and accessible system focused on real-time IoT sensing and irrigation management.

5.1.3 Use of a Digital Twin for Water Efficient Management in a Processing Tomato Commercial Farm [24]

Overview

Millan et al. present a digital twin for water-efficient management in a processing tomato commercial farm. The paper is relevant because it studies a real agricultural production context and focuses directly on water management. It shows how digital twin concepts can be applied outside controlled industrial environments and inside an operational farm.

Architecture / Methodology

The methodology links a physical commercial tomato farm with a digital representation used for water-management decisions. The physical side corresponds to the crop, soil, climate, and irrigation infrastructure. Data from the farm are used to describe the current state of the production system.

The virtual side is used to reason about crop water needs and irrigation scheduling. In this type of system, the digital twin must represent the farm state at a level that is useful for daily water management. This means that the model should connect crop status, soil water availability, and irrigation decisions.

The commercial-farm context is important. Many digital twin systems are validated only as prototypes, but this paper focuses on a real production environment. The methodology therefore emphasizes practical water management, not only theoretical modeling.

For this thesis, the paper shows the importance of moving from laboratory monitoring to field-oriented irrigation management. A practical smart irrigation digital twin should be able to receive farm data, update the virtual state, and support decisions that can be used in real irrigation operations.

Evaluation and Results

The paper is a commercial-farm case study. The available verified metadata confirms the application to water-efficient management in a processing tomato farm, but it does not provide detailed numerical values for water saving, yield, or prediction error. Therefore, this review treats the evaluation as field-oriented qualitative validation.

Limitations and Relevance to This Project

The main limitation is that exact performance metrics and implementation details are not available in the verified metadata used here. The paper remains relevant because it shows the importance of validating digital twin irrigation systems in real commercial farms. The gap is to document an end-to-end architecture that can be reproduced from sensing to virtual state, decision support, and irrigation management.

5.2 Digital Twins with Machine Learning and Control

5.2.1 Digital Twin Enhanced with Machine Learning Algorithms for Irrigation Management Using Sensor Data [25]

Overview

Tancredi et al. propose a digital twin enhanced with machine-learning algorithms for irrigation management using sensor data. The paper addresses the need to transform raw sensor measurements into irrigation decisions. It is relevant because it explicitly connects sensor data, a digital twin layer, and machine-learning decision support.

Architecture / Methodology

The system uses sensor data as the input to a digital twin environment. The first methodological step is therefore data acquisition from the agricultural context. These measurements describe the current state of the irrigation environment and provide the basis for later prediction.

The digital twin layer organizes the sensor information into a virtual representation. This representation gives structure to the data and allows the system to reason about irrigation state rather than treating each measurement separately. The virtual layer is therefore the bridge between raw sensing and decision support.

Machine-learning algorithms are then applied to the sensor-based digital twin data. Their role is to support irrigation management decisions, such as identifying whether irrigation is needed or estimating the suitable action. This makes the methodology different from threshold-based automation because the decision can be learned from data patterns.

The workflow can be summarized as sensor acquisition, digital twin state organization, machine-learning analysis, and irrigation management decision. This is directly relevant to a smart irrigation digital twin because it shows how AI can be embedded into the decision layer.

Evaluation and Results

The paper evaluates machine-learning models for irrigation management and reports the use of classification-style metrics such as accuracy, precision, recall, F1-score, or confusion-matrix analysis. Exact numerical metric values are not available in the verified metadata used for this review. The evaluation is therefore described as algorithmic validation of a machine-learning-enhanced digital twin.

Limitations and Relevance to This Project

The limitation is that the available metadata does not provide enough detail about deployment scale, long-term robustness, or field validation. The paper is relevant because it shows how machine learning can be embedded into a digital twin for irrigation decisions. The remaining gap is to connect the learning layer to practical real-time monitoring, user-facing decision support, and irrigation control.

5.2.2 Digital Twin-Enabled Intelligent Irrigation-Drainage System for Precision Water-Salt Management in Saline Agroecosystems [26]

Overview

Qin et al. propose a digital twin-enabled intelligent irrigation-drainage system for precision water-salt management in saline agroecosystems. The paper addresses a more complex agricultural water problem than simple irrigation because it considers both water and salinity dynamics. This is relevant because irrigation decisions can affect soil salinity, crop stress, and long-term sustainability.

Architecture / Methodology

The proposed system links an irrigation-drainage physical environment with a digital twin for water-salt management. The physical side includes the saline agroecosystem, where irrigation and drainage decisions affect both water availability and salt movement. This makes the problem more complex than ordinary irrigation scheduling.

The data acquisition layer collects information from the agroecosystem. These data are used to represent the current water-salt state in the virtual layer. The digital twin therefore acts as a structured model of both irrigation conditions and salinity-related constraints.

The intelligent decision layer uses the virtual water-salt state to support irrigation and drainage decisions. The objective is not only to apply enough water, but also to manage salinity risks. This creates a more advanced decision problem because irrigation can improve water availability while also influencing salt transport.

The methodology is relevant because it shows how agricultural digital twins can include environmental state variables beyond soil moisture. For a practical first system, the main lesson is that the architecture should be extensible so additional agronomic variables can later be integrated.

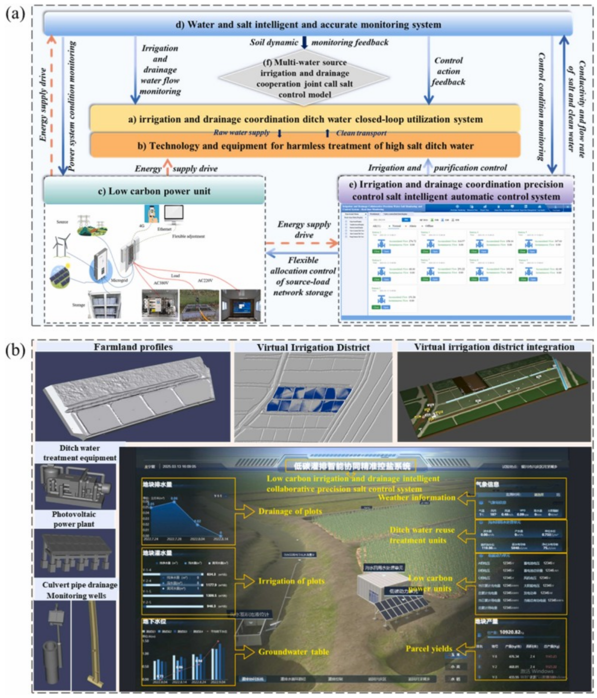

Figure 5.2: Digital twin-enabled irrigation-drainage framework for precision water-salt management, adapted from [26]. Source: https://doi.org/10.1016/j.agwat.2025.109957.

Evaluation and Results

The paper evaluates the proposed intelligent irrigation-drainage system in the context of saline agroecosystems. The available verified metadata confirms the concrete digital twin system and its water-salt management purpose, but it does not provide detailed numeric values for salinity reduction, water saving, or crop response. Therefore, the validation is treated as application-level qualitative validation in this review.

Limitations and Relevance to This Project

The limitation is the high agronomic and environmental specificity of saline agroecosystems. Such systems may require more complex sensors, models, and calibration than a general smart irrigation prototype. Its relevance is that it shows how digital twins can extend irrigation management beyond water quantity toward soil and environmental state. The gap remains the development of a practical system that begins with core sensing, virtual monitoring, and irrigation decision support before adding more complex agronomic models.

5.3 Digital Twins for Collective Irrigation Management

5.3.1 Towards Collective Intelligence in Agriculture: Deep Reinforcement Learning and Digital Twins for Efficient Management of Collective Irrigation Water Distribution Systems [27]

Overview

El Hachimi et al. study digital twins and deep reinforcement learning for the management of collective irrigation water distribution systems. The paper addresses a water-distribution problem where irrigation decisions affect multiple users or plots. It is relevant because it moves from individual field control to coordinated irrigation management.

Architecture / Methodology

The methodology combines a digital twin representation of a collective irrigation distribution system with a deep reinforcement-learning decision layer. The physical system is not a single plot or greenhouse. It is a shared water-distribution infrastructure where several demands and constraints must be coordinated.

The virtual layer represents the state of the irrigation distribution system. This state can include water availability, delivery constraints, and demand conditions. By modeling the collective system, the digital twin provides an environment where management strategies can be evaluated.

The reinforcement-learning component learns policies for efficient water distribution. Instead of applying fixed rules, the learning agent explores decisions and improves its policy according to performance objectives. This makes the methodology suitable for complex systems where the best action depends on the state of several connected components.

For this thesis, the paper is useful because it shows the future direction of irrigation digital twins: moving from local decision support to coordinated water management. However, such systems require more data and infrastructure than a single smart irrigation prototype.

Evaluation and Results

The work evaluates the combination of digital twins and deep reinforcement learning for collective irrigation management. The verified metadata confirms the use of this approach for efficient management of irrigation water distribution systems, but it does not expose exact quantitative results in the material consulted. The evaluation is therefore described as algorithmic and system-level validation.

Limitations and Relevance to This Project

The limitation is that collective irrigation distribution is more complex than a single smart irrigation unit and may require institutional, hydraulic, and multi-user data that are not always available. Its relevance is that it shows the future potential of digital twins for coordinated irrigation policies. The gap for this project is to focus first on practical real-time sensing, local virtual representation, and irrigation decision support before scaling to collective management.

5.4 Discussion

The selected agriculture and irrigation digital twin papers show that the field is moving from simple monitoring toward synchronized decision-support systems. Bellvert et al. provide the strongest verified quantitative evidence, with satellite assimilation, soil-water balance simulation, and daily irrigation prescriptions. Manocha et al. and Tancredi et al. show how IoT sensing and machine learning can support irrigation digital twins. Qin et al. expand the problem toward water-salt management, while El Hachimi et al. explore collective irrigation distribution using reinforcement learning. Millan et al. demonstrate the importance of commercial farm validation. Together, these papers show that irrigation digital twins are feasible, but they also show that many systems remain specific to one crop, one decision-support platform, one type of model, or one deployment context.

Table 5.1: Comparison of selected agricultural and irrigation digital twin papers

Paper	Agricultural context	Digital twin components	Data synchronization	AI / simulation model	Irrigation decision support	Validation	Limitation
Manocha et al. [22]	Smart irrigation	IoT layer and virtual irrigation representation	Sensor-driven update	Digital twin-inspired irrigation optimization	Water-utilization decisions	System validation	Limited public detail on metrics and deployment scale
Bellvert et al. [23]	Vineyard irrigation	DSS, IoT, Sentinel-2 assimilation, soil-water balance	Sensor, weather, and satellite inputs	Soil-water balance simulation and fAPAR assimilation	Daily irrigation prescriptions	Quantitative vineyard evaluation	Crop and platform specific
Millan et al. [24]	Processing tomato commercial farm	Farm digital twin for water management	Field and irrigation data	Water-management model	Efficient irrigation management	Commercial-farm case study	Detailed metrics not available in verified metadata
Tancredi et al. [25]	Sensor-based irrigation	Digital twin with ML layer	Sensor-data input	Machine-learning algorithms	Irrigation management decisions	Algorithmic validation	Limited detail on field deployment
Qin et al. [26]	Saline agroecosystems	Irrigation-drainage twin and water-salt state	Agroecosystem data input	Intelligent water-salt management model	Irrigation and drainage decisions	Application validation	High domain specificity and model complexity

Paper	Agricultural context	Digital twin components	Data synchronization	AI / simulation model	Irrigation decision support	Validation	Limitation
El Hachimi et al. [27]	Collective irrigation distribution	Digital twin and deep reinforcement learning	Distribution-system state representation	Deep reinforcement-learning policy	Collective water distribution management	System and algorithmic validation	Requires multi-user and infrastructure data

The main gaps are practical integration, reproducibility, and generality. Several works demonstrate advanced models, but many depend on specific platforms, specific crops, specialized data sources, or complex calibration. A practical smart irrigation digital twin still needs a clear end-to-end architecture that joins IoT sensing, live state monitoring, virtual representation, prediction, recommendation, and irrigation management in a form that can be implemented and extended.

5.5 State-of-the-Art Synthesis and Research Gap

Industrial digital twins already show strong methods for connecting a physical asset to a virtual representation. They demonstrate real-time data acquisition, service layers, visualization, prediction, and control. Their main contribution to this thesis is architectural: a digital twin must be a synchronized system, not only a dashboard or a simulation model.

Smart irrigation systems already show that field sensing and automated irrigation are technically feasible. Wireless sensor networks, GPRS, MQTT, and IoT platforms can collect soil or environmental data and transmit it to a controller. Machine-learning approaches can also support irrigation decisions. However, many smart irrigation systems stop at monitoring, rule-based automation, or isolated prediction.

Digital twins in agriculture and irrigation bring these two directions together, but important limitations remain. Existing works are often crop-specific, platform-specific, or focused on one narrow decision problem. Some works provide strong modeling but limited implementation detail. Others provide prototype architectures but limited field validation or quantitative results. There is still a need for practical systems that connect sensing, synchronization, virtual representation, decision support, and irrigation management in one coherent workflow.

This project is justified by this gap. The target is not only to predict irrigation or display sensor readings. The gap addressed is the integration of IoT sensing, real-time monitoring, a virtual representation of the irrigation context, decision support, and irrigation management in a practical smart irrigation digital twin system. Such an integration can help transform smart irrigation from isolated sensing and control components into a coherent digital twin workflow.

General Conclusion

This Master Diploma report presented the scientific study of Agrio, a Digital Twin-based intelligent irrigation platform for precision agriculture. The work focused on the background, state of the art, and synthesis required to understand how Digital Twins can support smart irrigation and agricultural water management.

The background chapters showed that smart irrigation is motivated by water scarcity, climate variability, and the need to improve irrigation decisions. Precision agriculture, IoT, wireless communication, artificial intelligence, simulation, and human-in-the-loop decision support form the main technological foundation of this field. Digital Twin technology extends classical monitoring by connecting the physical state of an agricultural system with a virtual representation able to support prediction, simulation, recommendation, and feedback.

The state-of-the-art review showed that Digital Twins are increasingly studied in agriculture but remain difficult to deploy due to data heterogeneity, biological complexity, cost, connectivity, calibration, and limited long-term validation. The literature also shows that many systems focus on monitoring or decision support but do not always implement a complete sensing-to-actuation loop.

The synthesis and comparative analysis positioned Agrio as an integrated project that combines concepts from smart irrigation, agricultural Digital Twins, IoT, machine learning, simulation, human validation, and actuation. This positioning does not claim that Agrio is a fully mature field-deployed Digital Twin. Instead, it identifies the scientific gap that the engineering prototype addresses: building a coherent bridge between research concepts and an operational cyber-physical irrigation workflow.

Future scientific work should extend the review with new field-validated systems, compare additional machine learning and optimization approaches, study agronomic calibration in greater depth, and evaluate Digital Twin maturity using long-term agricultural deployments.

Bibliography

[1] International Society of Precision Agriculture, "Precision agriculture definition," 2024. [Online]. Available: https://www.ispag.org/resources/definition

[2] L. Garcia, L. Parra, J. M. Jimenez, J. Lloret, and P. Lorenz, "Iot-based smart irrigation systems: An overview on the recent trends on sensors and iot systems for irrigation in precision agriculture," *Sensors*, 2020.

[3] K. Obaideen *et al.*, "An overview of smart irrigation systems using iot," *Energy Nexus*, 2022.

[4] A. Fuller, Z. Fan, C. Day, and C. Barlow, "Digital twin: Enabling technologies, challenges and open research," *IEEE Access*, vol. 8, pp. 108 952–108 971, 2020. [Online]. Available: https://doi.org/10.1109/ACCESS.2020.2998358

[5] W. Kritzinger, M. Karner, G. Traar, J. Henjes, and W. Sihn, "Digital twin in manufacturing: A categorical literature review and classification," *IFAC-PapersOnLine*, vol. 51, no. 11, pp. 1016–1022, 2018. [Online]. Available: https://doi.org/10.1016/j.ifacol.2018.08.474

[6] J. Gubbi, R. Buyya, S. Marusic, and M. Palaniswami, "Internet of things (iot): A vision, architectural elements, and future directions," *Future Generation Computer Systems*, vol. 29, no. 7, pp. 1645–1660, 2013. [Online]. Available: https://arxiv.org/abs/1207.0203

[7] J. McCarthy, M. L. Minsky, N. Rochester, and C. E. Shannon, "A proposal for the dartmouth summer research project on artificial intelligence," 1955. [Online]. Available: https://www-formal.stanford.edu/jmc/history/dartmouth/dartmouth.html

[8] L. Breiman, "Random forests," *Machine Learning*, 2001.

[9] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, 2015. [Online]. Available: https://www.nature.com/articles/nature14539

[10] R. Bommasani, D. A. Hudson, E. Adeli, R. Altman, S. Arora, S. von Arx, M. S. Bernstein, J. Bohg, A. Bosselut, E. Brunskill *et al.*, "On the opportunities and risks of foundation models," 2021. [Online]. Available: https://arxiv.org/abs/2108.07258

[11] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," 2017. [Online]. Available: https://arxiv.org/abs/1706.03762

[12] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, S. Agarwal, A. Herbert-Voss, G. Krueger, T. Henighan, R. Child, A. Ramesh, D. M. Ziegler, J. Wu, C. Winter, C. Hesse, M. Chen, E. Sigler, M. Litwin, S. Gray, B. Chess, J. Clark, C. Berner, S. McCandlish, A. Radford, I. Sutskever, and D. Amodei, "Language models are few-shot learners," in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 1877–1901. [Online]. Available: https://arxiv.org/abs/2005.14165

[13] M. Wooldridge and N. R. Jennings, "Intelligent agents: Theory and practice," *The Knowledge Engineering Review*, vol. 10, no. 2, pp. 115–152, 1995.

[14] T. H.-J. Uhlemann, C. Lehmann, and R. Steinhilper, "The digital twin: Realizing the cyber-physical production system for industry 4.0," in *Procedia CIRP*, vol. 61, 2017, pp. 335–340. [Online]. Available: https://doi.org/10.1016/j.procir.2016.11.152

[15] G. Schroeder, C. Steinmetz, C. E. Pereira, I. Muller, N. Garcia, D. Espindola, and R. Rodrigues, "Visualising the digital twin using web services and augmented reality," in *2016 IEEE 14th International Conference on Industrial Informatics (INDIN)*, 2016, pp. 522–527. [Online]. Available: https://doi.org/10.1109/INDIN.2016.7819217

[16] C. Zhuang, J. Liu, and H. Xiong, "Digital twin-based smart production management and control framework for the complex product assembly shop-floor," *The International Journal of Advanced Manufacturing Technology*, vol. 96, no. 1–4, pp. 1149–1163, 2018. [Online]. Available: https://doi.org/10.1007/s00170-018-1617-6

[17] V. Damjanovic-Behrendt and W. Behrendt, "An open source approach to the design and implementation of digital twins for smart manufacturing," *International Journal of Computer Integrated Manufacturing*, vol. 32, no. 4–5, pp. 366–384, 2019. [Online]. Available: https://doi.org/10.1080/0951192X.2019.1599436

[18] Y. Kim, R. G. Evans, and W. M. Iversen, "Remote sensing and control of an irrigation system using a distributed wireless sensor network," *IEEE Transactions on Instrumentation and Measurement*, vol. 57, no. 7, pp. 1379–1387, 2008. [Online]. Available: https://doi.org/10.1109/TIM.2008.917198

[19] J. Gutierrez, J. F. Villa-Medina, A. Nieto-Garibay, and M. A. Porta-Gandara, "Automated irrigation system using a wireless sensor network and gprs module," *IEEE Transactions on Instrumentation and Measurement*, vol. 63,

no. 1, pp. 166–176, 2014. [Online]. Available: https://doi.org/10.1109/TIM.2013.2276487

[20] R. K. Kodali and B. S. Sarjerao, “A low cost smart irrigation system using mqtt protocol,” in *2017 IEEE Region 10 Symposium (TENSYMP)*, 2017, pp. 1–5. [Online]. Available: https://doi.org/10.1109/TENCONSpring.2017.8070095

[21] A. Goap, D. Sharma, A. K. Shukla, and C. R. Krishna, “An iot based smart irrigation management system using machine learning and open source technologies,” *Computers and Electronics in Agriculture*, vol. 155, pp. 41–49, 2018. [Online]. Available: https://doi.org/10.1016/j.compag.2018.09.040

[22] A. Manocha, S. K. Sood, and M. Bhatia, “Iot-digital twin-inspired smart irrigation approach for optimal water utilization,” *Sustainable Computing: Informatics and Systems*, vol. 41, p. 100947, 2024. [Online]. Available: https://doi.org/10.1016/j.suscom.2023.100947

[23] J. Bellvert, A. Pelecha, M. Pamies-Sans, J. Virgili, M. Torres, and J. Casadesus, “Assimilation of sentinel-2 biophysical variables into a digital twin for the automated irrigation scheduling of a vineyard,” *Water*, vol. 15, no. 14, p. 2506, 2023. [Online]. Available: https://doi.org/10.3390/w15142506

[24] S. Millan, C. Montesinos, J. Casadesus, J. M. Vadillo, and C. Campillo, “Use of a digital twin for water efficient management in a processing tomato commercial farm,” *Agronomy*, vol. 15, no. 9, p. 2132, 2025. [Online]. Available: https://doi.org/10.3390/agronomy15092132

[25] G. P. C. Tancredi, L. Preite, and G. Vignali, “Digital twin enhanced with machine learning algorithms for irrigation management using sensor data,” *Procedia Computer Science*, vol. 253, pp. 2419–2427, 2025. [Online]. Available: https://doi.org/10.1016/j.procs.2025.01.302

[26] M. Qin, C. Zhang, G. Ma, Y. Ma, X. Feng, P. Li, and K. Feng, “Digital twin-enabled intelligent irrigation-drainage system for precision water-salt management in saline agroecosystems,” *Agricultural Water Management*, vol. 322, p. 109957, 2025. [Online]. Available: https://doi.org/10.1016/j.agwat.2025.109957

[27] C. El Hachimi, S. Belaqziz, S. Khabba, H. Karjoun, M. H. Kharrou, B. Ait Hssaine, S. Er-Raki, A. Daccache, A. Chehbouni, and Q. Weng, “Towards collective intelligence in agriculture: Deep reinforcement learning and digital twins for efficient management of collective irrigation water distribution systems,” *Energy Nexus*, vol. 20, p. 100599, 2025. [Online]. Available: https://doi.org/10.1016/j.nexus.2025.100599

[28] A. Subeesh and N. Chauhan, “Agricultural digital twin for smart farming: A review,” *Green Technologies and Sustainability*, 2026.

[29] R. Zhang, H. Zhu, Q. Chang, and Q. Mao, "A comprehensive review of digital twins technology in agriculture," *Agriculture*, 2025.

[30] R. Gund, C. M. Badgujar, S. Samiappan, and S. Jagadamma, "Application of digital twin technology in smart agriculture: A bibliometric review," *Agriculture*, 2025.

[31] M. Awais, X. Wang, S. Hussain, F. Aziz, and M. Q. Mahmood, "Advancing precision agriculture through digital twins and smart farming technologies: A review," *AgriEngineering*, 2025.

[32] M. Tsaousidis, T. Kalampokas, E. Vrochidou, and G. A. Papakostas, "Ai-enabled digital twins in agriculture," *AI*, 2026.

[33] R. Ahsen *et al.*, "Harnessing digital twins for sustainable agricultural water management: A systematic review," *Applied Sciences*, 2025.

[34] M. Del-Coco, M. Leo, and P. Carcagni, "Machine learning for smart irrigation in agriculture: How far along are we?" *Information*, 2024.

[35] A. Morchid, Z. Said, and H. Tairi, "Innovative applications of internet of things and machine learning in sustainable agricultural irrigation management: Benefits and challenges," *Smart Agricultural Technology*, 2026.

[36] C. Balaceanu, G. Suciu, and I. Marcu, "Libelium-based iot monitoring solution for precision agriculture," *Journal of E-Technology*, 2019.

[37] Liu *et al.*, "Digital twin technology for cotton canopy development: A water-stress perspective," *Smart Agricultural Technology*, 2025.

[38] D. C. Montgomery, E. A. Peck, and G. G. Vining, *Introduction to Linear Regression Analysis*. Wiley, 2021.

[39] FastAPI Documentation, "Fastapi framework and automatic openapi documentation," 2026.

[40] Docker Documentation, "Docker compose overview," 2026.

[41] HiveMQ, "Mqtt essentials and publish/subscribe messaging," 2026.

[42] Open-Meteo Documentation, "Weather forecast api," 2026.

[43] Libelium, "Smart agriculture iot vertical kit and plug & sense sensing platform," 2017.

[44] A. Z. Babar and O. B. Akan, "Sustainable and precision agriculture with the internet of everything," 2024. [Online]. Available: https://arxiv.org/abs/2404.06341

[45] A. Risheh, A. Jalili, and E. Nazerfard, "Smart irrigation iot solution using transfer learning for neural networks," 2020. [Online]. Available: https://arxiv.org/abs/2009.12747

[46] M. E. Karar, M. F. Al-Rasheed, A. F. Al-Rasheed, and O. Reyad, "Iot and neural network-based water pumping control system for smart irrigation," 2020. [Online]. Available: https://arxiv.org/abs/2005.04158

[47] A. Sharma, E. Kosasih, J. Zhang, A. Brintrup, and A. Calinescu, "Digital twins: State of the art theory and practice, challenges, and open research questions," 2020. [Online]. Available: https://arxiv.org/abs/2011.02833